SIMATS ENGINEERING

ASSESSMENT TOOL I – EXPERIMENTS

Course: Data Handling and Visualization	Code: DSA0603	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No : 192324285
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COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Experiments

Weightage: 20%

Code of Conduct:

I Surya JK / 192324285(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Understanding of Concepts 25%	Experimental Design 30%	Use of Tools 20%	Analysis of Results 15%	Documents 10%
Q1					
Q2					
Q3					
Q4					
Q5					
Total					

DSA06 – DATA HANDLING AND VISUALIZATION CO3_AT1 – Assessment Tool: EXPERIMENT (with Model Answers)

Question 1 – Distribution Diagnosis using ECDF and Q-Q Plot

Scenario: Response time (ms) of an automated fault-detection system, 20 test runs. The team claims the distribution is approximately normal and wants to use mean ± SD for monitoring.

Dataset (n=20): 118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162, 171, 184, 205, 249, 318

Answer

Summary statistics: Mean ≈ 159.35 ms, Median = 142.5 ms, Sample SD ≈ 49.0 ms, Q1 = 130, Q3 = 166.5, IQR = 36.5.

1. ECDF and Q-Q Plot

The ECDF, F(x), plots the cumulative proportion of observations ≤ x against x. For this data the ECDF rises steeply between 118–170 ms (where 15 of the 20 values are concentrated) and then flattens into a long, sparse right tail extending out to 205, 249, and 318 ms — a visual signature of positive (right) skew rather than a symmetric S-curve.

The normal Q-Q plot plots sorted sample quantiles against theoretical normal quantiles. For the bulk of the data (roughly the first 15–16 points) the points would lie close to the reference line, but the upper tail values (184, 205, 249, 318) would bow sharply upward and away from the line — the classic Q-Q signature of a heavy right tail / positive skew.

2. Evaluation of the normality claim

Both diagnostics reject the normality claim. The ECDF is not the smooth symmetric S-shape expected under normality (it is compressed on the left and stretched on the right), and the Q-Q plot's upward-curving tail confirms departure from normality specifically in the upper tail. The mean (159.35) also exceeds the median (142.5) by a wide margin — a hallmark of right-skewed data — whereas mean ≈ median is expected under normality.

3. Evidence of skewness / extreme observations

Using the 1.5×IQR rule: Lower fence = Q1 − 1.5×IQR = 130 − 54.75 = 75.25; Upper fence = Q3 + 1.5×IQR = 166.5 + 54.75 = 221.25.

249 ms and 318 ms both exceed the upper fence and are statistical outliers (10% of the sample). 205 ms is borderline/elevated. No values fall below the lower fence, confirming the skew is one-sided (right/positive skew) rather than symmetric spread.

4. Recommendation

Because the data is right-skewed with genuine outliers, mean ± SD control limits are inappropriate — they assume symmetry and would under-estimate the upper limit (mean + 2SD ≈ 257 ms would flag 318 ms but miss the broader tail behaviour) while producing a meaningless negative-side lower limit. Percentile-based limits (e.g., the 90th/95th/99th percentile of the empirical distribution, or a limit based on the median and IQR such as Q3 + 1.5×IQR ≈ 221 ms) are recommended, since they make no symmetry assumption and directly reflect the actual shape of the response-time distribution. A log-transform prior to using mean±SD is an acceptable alternative if a parametric limit is still desired.

Question 2 – Comparing Many Distributions and Bean Plots

Scenario: Chip fabrication cycle time (minutes) for four production lines. Management wants to identify differences in variability and distribution shape.

Line	Cycle times (minutes)
L1	42,44,45,46,47,48,49,50,51,53
L2	38,41,44,47,50,53,56,59,62,65
L3	45,46,46,47,47,48,48,49,49,50
L4	40,41,42,43,44,45,48,55,61,72

Answer

Descriptive statistics (n=10 each):

Line	Mean	Range	Shape
L1	47.5	11 (42–53)	Fairly symmetric, moderate spread
L2	51.5	27 (38–65)	Uniform/linear spread — evenly spaced values, no concentration
L3	47.5	5 (45–50)	Tightly concentrated, near-symmetric
L4	49.1	32 (40–72)	Right-skewed with high-end outliers (61, 72)

1–2. Visualization

A comparative visualization places all four lines side by side along a common vertical (value) axis, e.g., box plots or bean plots for L1–L4 positioned along the horizontal category axis. A bean plot overlays: (a) a horizontal line for the mean, (b) a small 1-D scatter/rug of the individual points, and (c) a mirrored density (violin-style) curve showing where each line's cycle times are concentrated. This reveals L3's density as a single sharp narrow peak, L1's as a moderately narrow peak, L2's as a wide, flat, almost-rectangular density (values spread nearly uniformly), and L4's as a peak on the low end with a thin trailing bulge toward 55–72 (right-skew).

3. Greatest variability vs. most concentrated

L4 has the greatest variability (largest range, 32 minutes, and a right-skewed spread driven by two high values, 61 and 72, indicating occasional process disruptions). L3 has the most concentrated distribution (smallest range, 5 minutes, all ten values clustered tightly between 45–50, indicating a highly stable, well-controlled process).

4. Why comparing only means is insufficient

L1 and L3 share an identical mean (47.5 minutes), yet L1's range (11) is more than double L3's (5) — the mean alone completely hides this difference in process stability. Similarly, L2 and L4 have different shapes (L2 uniform, L4 right-skewed with outliers) despite comparable spread, so a viewer relying on means only would wrongly conclude the four lines perform similarly. Visualizing the full distribution exposes variability, skewness, and outliers that averages mask — information that is critical for a manufacturer deciding which line needs process-control intervention.

Question 3 – Proportion Visualization and Choice of Chart

Scenario: Preferred learning mode of 240 students across three departments. The dean wants comparison without exaggerating small differences.

Department	Online (%)	Blended (%)	Face-to-Face (%)
CSE	52	28	20
ECE	45	35	20
IT	30	40	30

Answer

1. Chart construction

Side-by-side (grouped/clustered) bar chart: for each department, three adjacent bars (Online, Blended, Face-to-Face) share a single common baseline (0%) on a shared y-axis, grouped by department.

100% stacked bar chart: one bar per department, each of fixed total height (100%), divided into three stacked segments in proportion to Online/Blended/Face-to-Face.

2. Which chart better supports comparing individual proportions, and why

The side-by-side bar chart is better for comparing individual proportions, because every bar starts from the same zero baseline. Per the Cleveland–McGill perceptual hierarchy, judging length/position along a common scale is the most accurate visual comparison humans can make. In a 100% stacked bar, only the bottom segment (Online) shares a common baseline; the Blended and Face-to-Face segments float at different heights across departments, forcing the viewer to compare segment lengths in isolation — a much harder and less accurate judgment. The side-by-side chart therefore reveals, for example, that CSE's Online share (52%) is clearly larger than IT's (30%) far more precisely than the stacked version does.

3. Why separate pie charts are less effective

Comparing several pie charts requires comparing angles (or areas of wedges) across physically separate circles — a perceptual task humans perform poorly and inconsistently. There is no shared baseline between pies, small percentage differences (e.g., 20% vs 20% vs 30% for Face-to-Face) are very hard to distinguish by eye, and the visual weight of a pie is dominated by its largest slice, making cross-department comparison of the smaller categories especially unreliable.

4. A design choice that could exaggerate small differences

Truncating or not starting the y-axis at zero (e.g., starting the bar chart's axis at 15% instead of 0%) would visually stretch small percentage differences between departments, making them appear far larger than they actually are. (Other examples: using 3-D tilted pie/bar effects, or inconsistent scales across side-by-side panels.)

Question 4 – Nested Proportions using Treemap and Sunburst

Scenario: E-commerce annual revenue across product categories and subcategories. The dashboard must show both category contribution and internal composition.

Category	Subcategory	Revenue (₹ lakh)
Electronics	Mobiles	24
Electronics	Laptops	18
Electronics	Accessories	8
Home	Furniture	14
Home	Appliances	10
Home	Decor	6
Fashion	Men	7
Fashion	Women	8
Fashion	Kids	5

Answer

1–2. Treemap and sunburst construction

Treemap: three top-level rectangles (Electronics, Home, Fashion), each subdivided into nested rectangles for its subcategories, with the area of every rectangle proportional to its revenue (₹ lakh).

Sunburst: an inner ring with three arcs (Electronics, Home, Fashion) sized by category revenue, and an outer ring with arcs for each subcategory, positioned radially within its parent category's angular span, arc length proportional to revenue.

3. Largest category and its share

Category totals: Electronics = 24+18+8 = 50 lakh; Home = 14+10+6 = 30 lakh; Fashion = 7+8+5 = 20 lakh. Sum of the table = ₹100 lakh (note: this is ₹1 crore based on the itemized data provided — slightly below the ₹1.2 crore headline figure in the scenario, so percentages below are computed against the ₹100 lakh sum of the given subcategories).

Electronics is the largest category at 50 lakh, i.e., 50 ÷ 100 × 100% = 50% of total revenue.

4. Treemap vs sunburst for nested proportions

The treemap uses area encoding and is generally more accurate for comparing magnitudes, since it makes efficient use of space and lets a viewer directly compare rectangle sizes (e.g., Mobiles vs Furniture) without needing to trace radial position. The sunburst uses arc length/angle and radial position, which makes the hierarchy (which subcategory belongs to which category) visually more explicit at a glance, but comparing two arcs at different radii or different rings is a harder perceptual task than comparing two rectangles, and outer-ring segments naturally get more physical space per unit value than inner-ring segments, which can bias perception.

5. A situation where area encoding could mislead

Because human perception of area is non-linear (we tend to under-estimate ratios of area, per Weber–Fechner-type perceptual scaling), a subcategory that is twice the revenue of another may not be perceived as "twice as big," so viewers can under-estimate real differences (e.g., Mobiles at 24 vs Decor at 6 — a 4× difference — may be perceived as noticeably less than 4×). Additionally, treemap layout algorithms can produce long, thin "sliver" rectangles for small values, which are hard to compare fairly to compact square rectangles of similar area, further distorting perceived importance.

Question 5 – Flow Proportions using Sankey Diagram / Parallel Sets

Scenario: 200 commuters followed from residential zone → transport mode → destination zone. The planning team wants dominant flows and possible bottlenecks.

Origin	Mode	Destination	Commuters
Residential	Bus	Central	35
Residential	Bus	Industrial	15
Residential	Train	Central	25
Residential	Train	Industrial	10
Suburban	Bus	Central	20
Suburban	Bus	Industrial	10
Suburban	Train	Central	30
Suburban	Train	Industrial	15
Rural	Bus	Central	12
Rural	Bus	Industrial	8
Rural	Train	Central	10
Rural	Train	Industrial	10

Answer

1–2. Sankey diagram and parallel sets

Sankey diagram: three vertical node columns — Origin (Residential, Suburban, Rural) → Mode (Bus, Train) → Destination (Central, Industrial) — connected by curved links whose width is proportional to the number of commuters on that path (total flow = 200, verified by summing the table).

Parallel sets: three parallel categorical axes (Origin, Mode, Destination) with ribbons connecting category values across axes, ribbon thickness again proportional to commuter count, allowing the same three-stage flow to be read as stacked segments at each axis.

3. Three largest flows and their percentages (of 200 total)

Rank	Flow	Commuters	% of total
1	Residential → Bus → Central	35	17.5%
2	Suburban → Train → Central	30	15.0%
3	Residential → Train → Central	25	12.5%

4. Sankey vs parallel sets

The Sankey diagram is stronger for showing flow magnitude, because link width is a direct, continuous quantitative encoding that lets viewers visually trace and compare the volume of each origin-to-destination path, including where flows merge or split (e.g., seeing that most flows converge on "Central"). Parallel sets are stronger for showing categorical relationships and set composition, since the vertical stacking at each axis makes it easy to see, for a given category (e.g., "Bus"), how it splits across the adjacent categories — but with many categories, parallel sets can become visually cluttered with numerous thin, hard-to-trace ribbons, whereas Sankey's free-form curved links usually remain easier to follow for tracing a single path end-to-end.

5. How poor link widths or ordering could mislead

If link widths are not drawn strictly proportional to the underlying commuter counts (e.g., a minimum link thickness is enforced for readability), small flows such as Rural→Bus→Industrial (8 commuters) would appear visually more significant than they truly are, hiding real bottleneck information. Likewise, ordering the nodes arbitrarily (e.g., alphabetically) rather than by flow volume can scatter the largest flows across the diagram, creating unnecessary crossing links that make the dominant Residential/Suburban → Central pattern much harder for planners to spot — the diagram would visually imply a more complex, evenly-distributed system than the data actually shows.